
Q1.Properties of Secure Hash Function

A **secure hash function** is used to convert input data into a fixed-size hash value securely.

Properties:

- **Deterministic** – Same input always produces the same hash output.
- **Fixed Output Length** – Produces hash of fixed size regardless of input size.
- **Pre-image Resistance** – It is difficult to find the original input from its hash.
- **Second Pre-image Resistance** – Hard to find a different input with the same hash as a given input.
- **Collision Resistance** – Difficult to find two different inputs that produce the same hash.
- **Fast Computation** – Hash value should be computed quickly.
- **Avalanche Effect** – A small change in input produces a large change in output.

Q2.List types of Authentication. Explain any one.

- **Password-based Authentication** – Uses username and password to verify identity.
- **Token-based Authentication** – Uses a physical or digital token (OTP, smart card) for access.
- **Biometric Authentication** – Uses unique physical or behavioral traits for identification.
- **Certificate-based Authentication** – Uses digital certificates to verify identity.
- **Multi-factor Authentication (MFA)** – Uses two or more authentication methods for higher security.

Explain: Biometric Authentication

Biometric authentication is a method of verifying a user's identity based on **unique physical or behavioral characteristics**.

Examples

- Fingerprint
- Face recognition
- Iris/Retina scan
- Voice recognition

Working

1. User provides biometric data (e.g., fingerprint).
2. System converts it into a **digital template**.
3. The template is compared with stored data.
4. If matched, access is granted.

Advantages

- High security and accuracy
- Difficult to forge
- No need to remember passwords

Disadvantages

- Expensive
- Privacy concerns
- Cannot be changed if compromised

Q3.What is need for message authrntication? List various techniques used for message authentication. Explain any one. / Discuss Message Authentication protocol(MAC).

Message authentication is required to ensure that the message received is **genuine, unaltered, and from a trusted sender**.

Needs:

- **Integrity** – Ensures message is not modified during transmission
- **Authentication** – Verifies the identity of the sender
- **Prevents forgery** – Avoids fake or altered messages
- **Security in communication** – Protects sensitive data

Techniques for Message Authentication

- Message Authentication Code (**MAC**)
- Hash Functions
- Digital Signatures
- HMAC (Hash-based Message Authentication Code)
- Encryption-based Authentication

[Explain: Message Authentication Code (MAC)]

Q4.Explain MD5 algorithm in detail.

MD5 Algorithm (Message Digest 5)

Definition

MD5 is a cryptographic hash function that takes input of any length and produces a fixed 128-bit (16-byte) hash value

- **Developed by Ronald Rivest (1991)**
- **Used for data integrity and authentication**

Working of MD5 Algorithm

Step 1: Append Padding Bits

- **Add padding bits so that message length becomes 64 bits less than multiple of 512**
  👉 (Shown in page 6 of your notes)

Step 2: Append Length

- **Append 64-bit representation of original message length**
  👉 Makes total length a multiple of 512

Step 3: Initialize MD Buffer

- **Use 4 buffers: A, B, C, D (each 32-bit)**
  👉 **Initial values are predefined**

Step 4: Process Message in Blocks

- **Divide message into 512-bit blocks**
- **Perform 64 operations in 4 rounds:**
  - **Round 1 → Function F**
  - **Round 2 → Function G**
  - **Round 3 → Function H**
  - **Round 4 → Function I**

👉 **Uses operations like AND, OR, XOR, NOT**

Step 5: Output

- **Final output is 128-bit message digest**
  👉 **Generated after all rounds**

Applications

- **Password storage**
- **File integrity checking**
- **Digital signatures**

⚡ Short Description (Easy to Remember)

👉 **“PAD – LENGTH – BUFFER – 4 ROUNDS – DIGEST”**

- **PAD → Add padding**
- **LENGTH → Add message length**
- **BUFFER → Initialize A, B, C, D**
- **4 ROUNDS → 64 operations (F, G, H, I)**
- **DIGEST → Get 128-bit output**

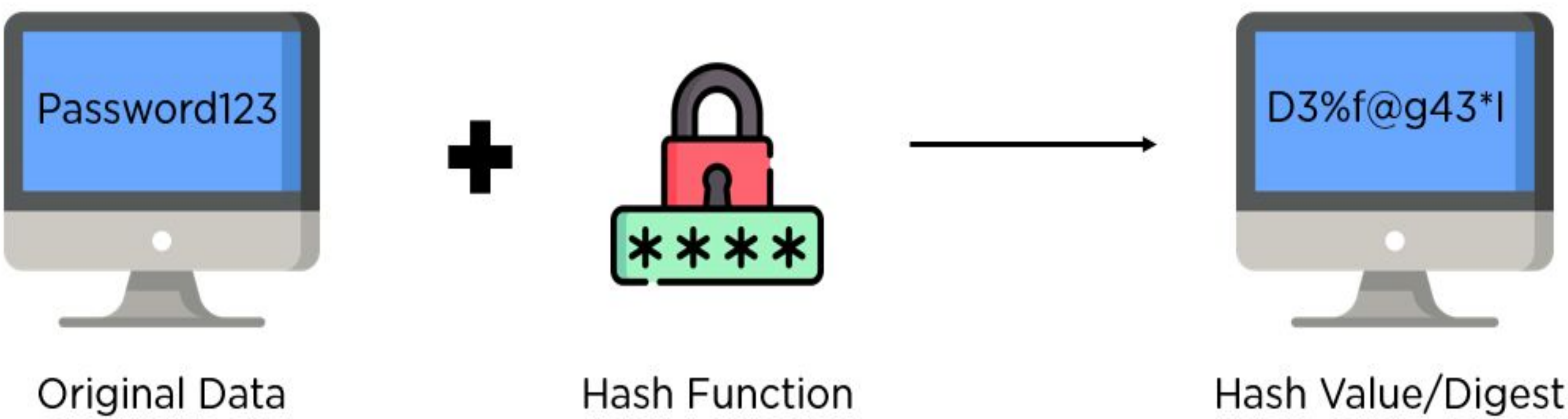
✔️ **This short trick helps you remember full answer in exam quickly** 👍

MD5

- MD5 algorithm stands for the **Message-Digest algorithm**.
- MD5 was developed in 1991 by **Ronald Rivest** as an improvement of MD4, with advanced security purposes.
- The output of MD5 (Digest size) is always **128 bits.**
- MD5 is primarily used to authenticate files.
- **MD5** is a cryptographic hash function algorithm that takes the message as input of any length and convert it into a fixed-length message of 16 bytes.

What Is Hashing?

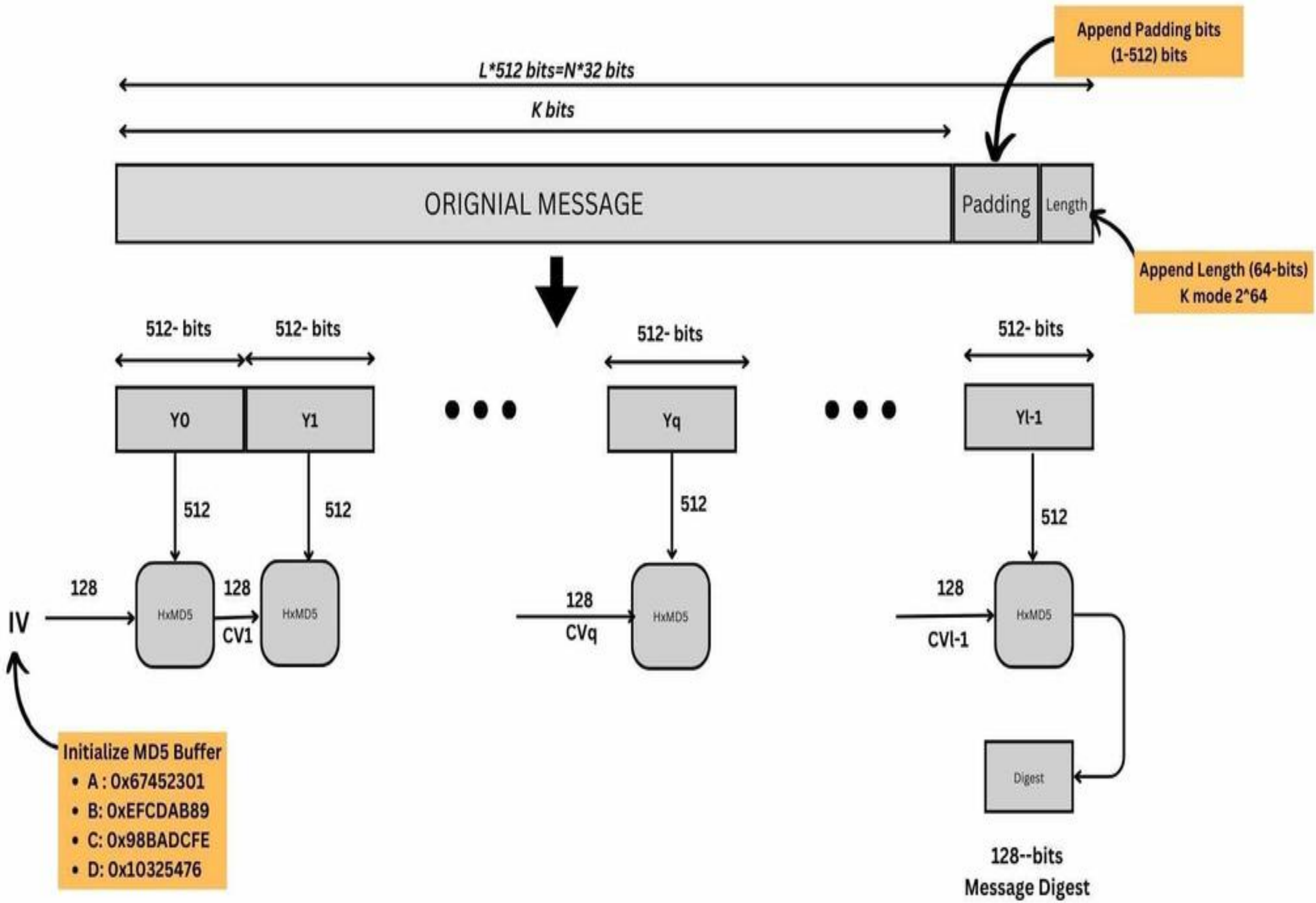
Hashing consists of converting a general string of information into an intricate piece of data.



Password Verification:

It is common to store user credentials of websites in a hashed format to prevent third parties from reading the passwords.

MD5 hashes are commonly used with smaller strings when storing passwords, credit card numbers or other sensitive data in databases such as the popular MySQL


Append Padding bits
(1-512) bits
L*512 bits=N*32 bits
K bits
ORIGNIAL MESSAGE
Padding
Length
Append Length (64-bits)
K mode 2^64
512- bits
512- bits
512- bits
512- bits
Y0
Y1
Yq
Yl-1
512
512
512
512
IV
128
HxMD5
128
CV1
HxMD5
128
CVq
HxMD5
128
CVl-1
HxMD5
Digest
Initialize MD5 Buffer
• A : 0x67452301
• B: 0xEFCDAB89
• C: 0x98BADCFE
• D: 0x10325476
128--bits
Message Digest

Working of the MD5 Algorithm

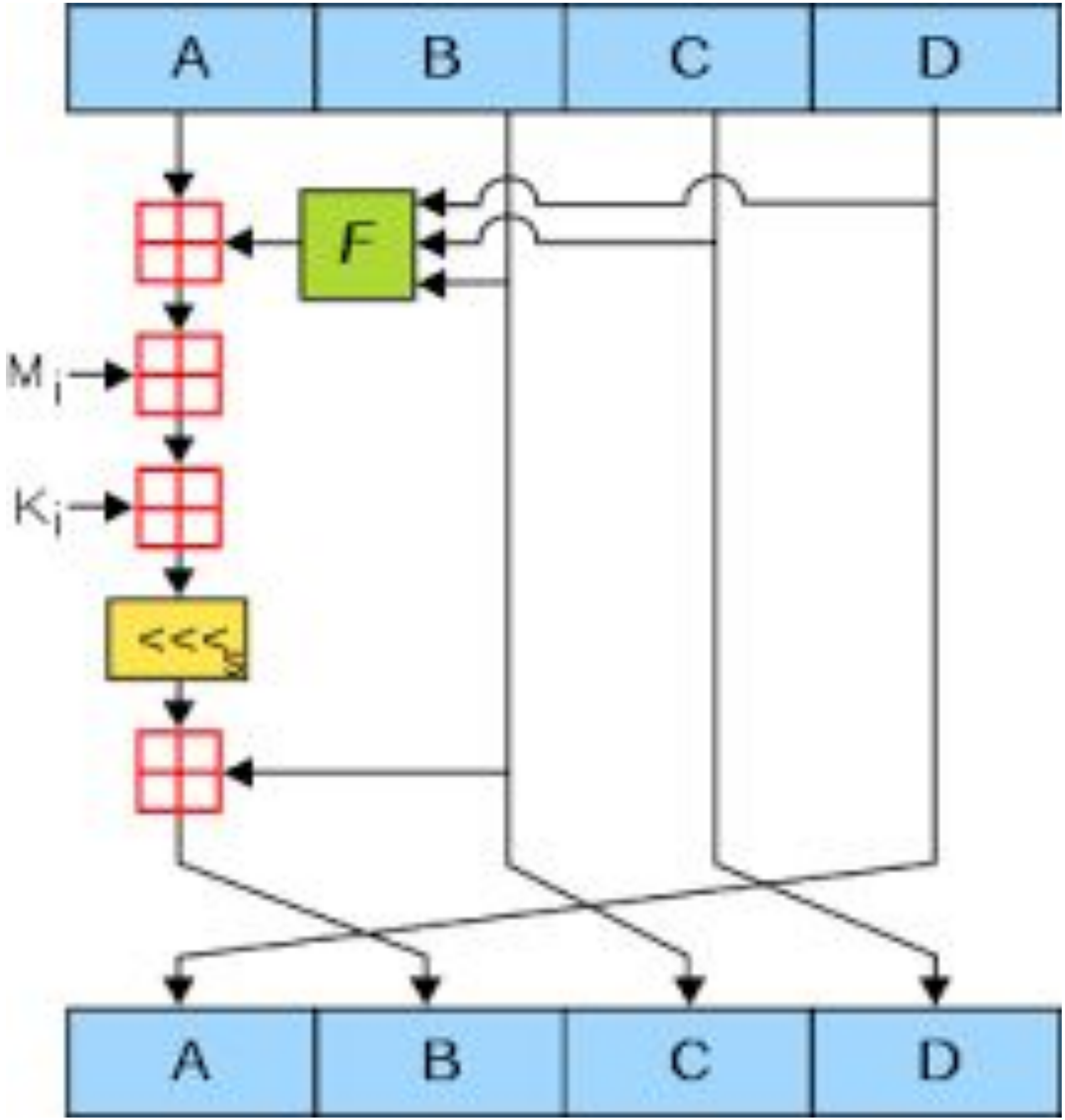
Append Padding Bits: In the first step, we add padding bits in the original message in such a way that the total length of the message is 64 bits less than the exact multiple of 512.

Suppose we are given a message of 1000 bits. Now we have to add padding bits to the original message. Here we will add 472 padding bits to the original message. After adding the padding bits the size of the original message/output of the first step will be 1472.

2. Append Length Bits:

In this step, we add the length bit in the output of the first step in such a way that the total number of the bits is the perfect multiple of 512.

Initialize MD buffer: Here, we use the 4 buffers i.e. A B, C, and D. The size of each buffer is 32 bits.



A
B
C
D
F
Mi
Ki
<<<s
A
B
C
D

4. Process Each 512-bit Block: This is the most important step of the MD5 algorithm.
Here, a total of 64 operations are performed in 4 rounds. In the 1st round, 16 operations will be performed, 2nd round 16 operations will be performed, 3rd round 16 operations will be performed, and in the 4th round, 16 operations will be performed.
We apply a different function on each round i.e. for the 1st round we apply the F function, for the 2nd G function, 3rd for the H function, and 4th for the I function.
We perform OR, AND, XOR, and NOT (basically these are logic gates) for calculating functions. We use 3 buffers for each function i.e. B, C, D.

After applying the function now we perform an operation on each block. For performing operations we need

add modulo 2^{32}

M[i] – 32 bit message.

K[i] – 32-bit constant.

$\lll n$ *– Left shift by n bits.*

$$A = B + ((A + F(B,C,D) + M[i] + K[i]) \lll n)$$

Now take input as initialize MD buffer i.e. A, B, C, D. Output of B will be fed in C, C will be fed into D, and D will be fed into J.
After doing this now we perform some operations to find the output for A.
In the first step, Outputs of B, C, and D are taken and then the function F is applied to them. We will add modulo 2^{32} bits for the output of this with A.
In the second step, we add the M[i] bit message with the output of the first step.
Then add 32 bits constant i.e. K[i] to the output of the second step.
At last, we do left shift operation by n (can be any value of n) and addition modulo by 2^{32}.

After all steps, the result of A will be fed into B.
Now same steps will be used for all functions G, H, and I.
After performing all 64 operations we will get our message digest.